Customer Segmentation and Prediction Project Report

1. Project Overview

This project analyzes customer demographics, income, purchase history, insurance product ownership, and customer preferences to understand customer behavior and identify meaningful customer segments.

The analysis uses Python for data cleaning, exploratory data analysis, feature engineering, K-Means clustering, and machine learning. The project also explores predicting future purchase behavior based on available purchase-history data.

The main objective is to identify different customer groups, understand their characteristics, evaluate predictive models, and recommend marketing strategies that can support customer engagement and increase product sales.

2. Dataset Summary

- Dataset Name: customer_segmentation_data.csv
- Rows after cleaning and feature engineering: 53,503
- Columns after feature engineering: 24
- Key Features:
    - Customer Details: Customer ID, Age, Gender, Marital Status, Education Level.
    - Customer Profile: Geographic Information, Occupation, Income Level.
    - Purchase Behavior: Behavioral Data, Purchase History.
    - Product Information: Insurance Products Owned, Policy Type.
    - Financial Information: Coverage Amount, Premium Amount.
    - Customer Preferences: Available preference-related fields and Segmentation Group.
    - Engineered Features: Age Group, Premium Income Ratio, Coverage Income Ratio, Premium Category.
- Duplicate Records: Duplicate rows were checked and removed during cleaning.
- Missing Data: Missing values were checked and handled during preprocessing; the cleaned dataset was reported to have zero missing values.

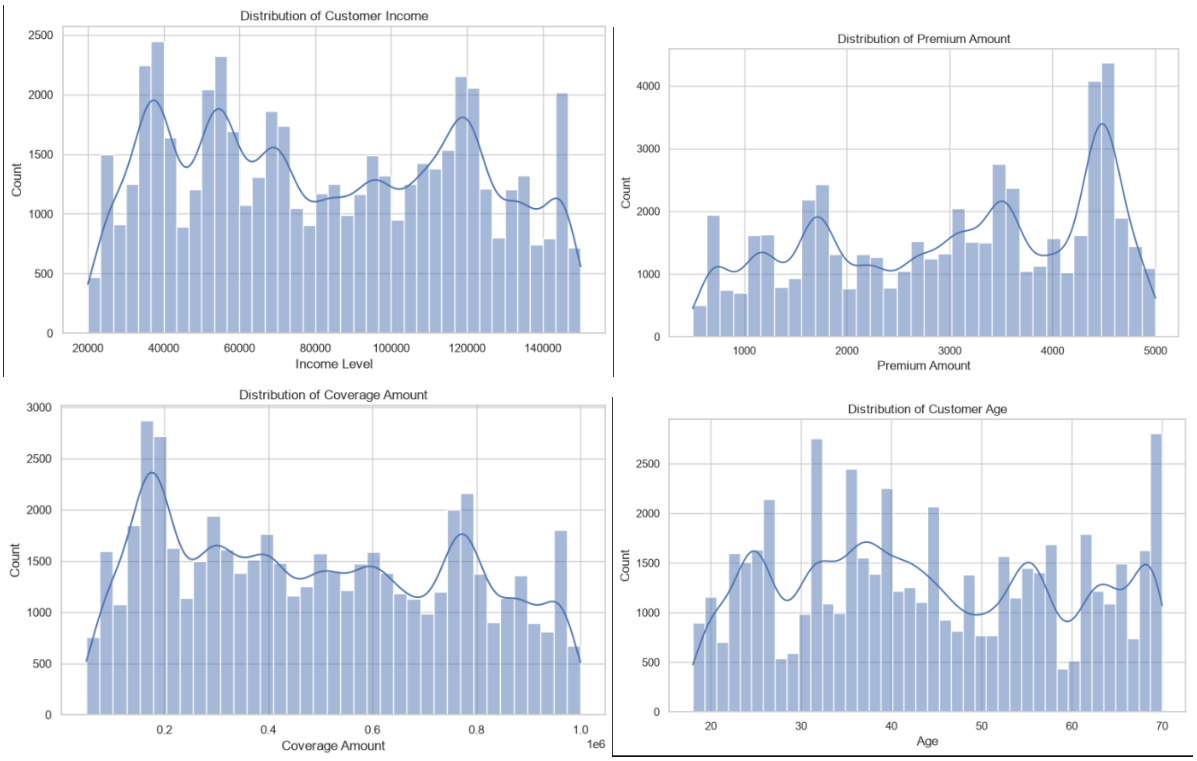
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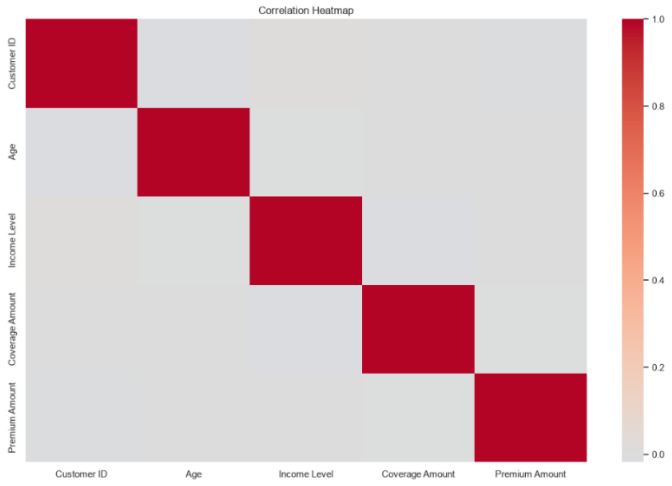
	Customer ID	Age	Gender	Marital Status	Education Level	Geographic Information	Occupation	Income Level	Behavioral Data	Purchase History	Interactions with Customer Service	Insurance Products Owned	Coverage Amount	Premium Amount	Policy Type	Cus Prefe
0	84966	23	Female	Married	Associate Degree	Mizoram	Entrepreneur	70541	policy5	04-10-2018	Phone	policy2	366603	2749	Group	
1	95568	26	Male	Widowed	Doctorate	Goa	Manager	54168	policy5	11-06-2018	Chat	policy1	780236	1966	Group	
2	10544	29	Female	Single	Associate Degree	Rajasthan	Entrepreneur	73899	policy5	06-05-2021	Email	policy3	773926	4413	Group	
3	77033	20	Male	Divorced	Bachelor's Degree	Sikkim	Entrepreneur	63381	policy5	09-02-2018	Chat	policy2	787815	4342	Family	
4	88160	25	Female	Separated	Bachelor's Degree	West Bengal	Manager	38794	policy1	09-10-2018	Chat	policy4	366506	1276	Family	

3. Exploratory Data Analysis Using Python

The dataset was processed and analyzed using Python, Pandas, Matplotlib, and Seaborn.

- Data Loading: Imported the customer dataset using Pandas.
- Initial Exploration: Used df.shape, df.head(), df.info(), and df.describe() to understand the dataset structure and numerical statistics.
- Data Cleaning: Checked and removed duplicate records and reviewed missing values.
- Missing-Value Handling: Processed missing data to prepare the dataset for analysis and modeling.
- Feature Engineering: Created Age Group, Premium Income Ratio, Coverage Income Ratio, and Premium Category.
- Age Analysis: Examined the distribution of customer ages.
- Income Analysis: Examined the distribution of customer income levels.
- Premium Analysis: Analyzed the distribution of premium amounts.
- Coverage Analysis: Examined customer coverage amounts.
- Customer Group Analysis: Reviewed the distribution of the existing Segmentation Group.
- Correlation Analysis: Used a correlation heatmap to examine relationships between numerical features.
- Data Quality Validation: Checked missing values, duplicate records, and data types after preprocessing.





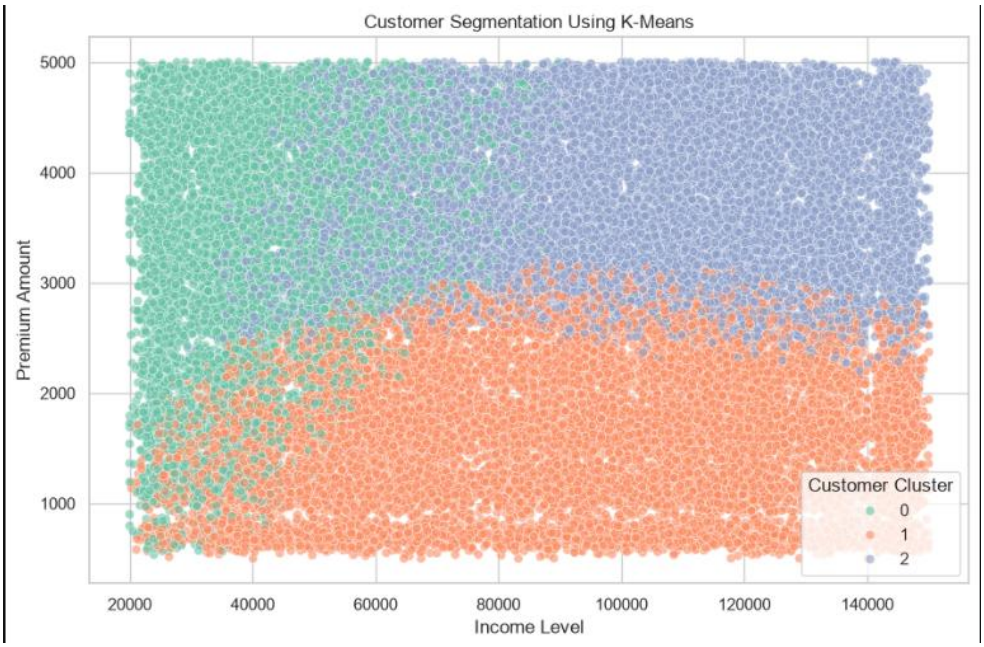
4. Customer Segmentation and Prediction Using Machine Learning

Machine learning techniques were used to group customers and explore customer prediction tasks.

4.1 Customer Segmentation Using K-Means

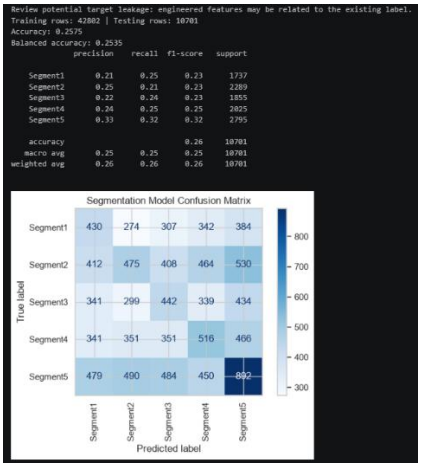
- Selected numerical features related to age, income, coverage, premium, and financial ratios.

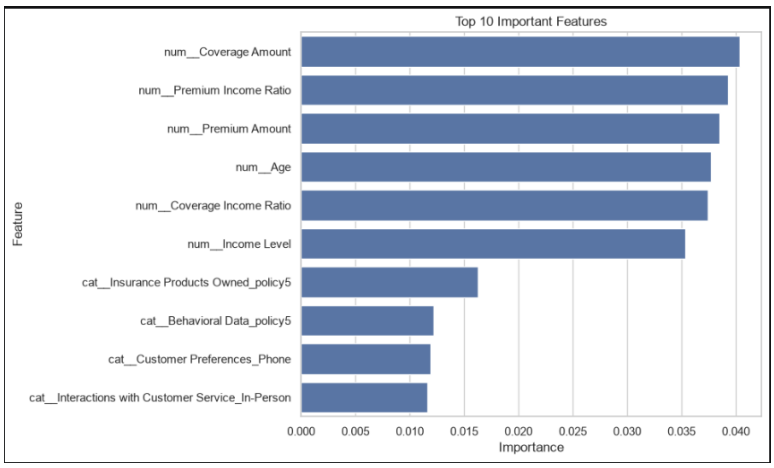
- Standardized the selected features before applying clustering.
- Applied the K-Means clustering algorithm with three clusters.
- Calculated the silhouette score to assess how well the clusters were separated.
- Summarized the characteristics of each cluster using group-level statistics.
- Visualized the customer clusters by comparing Income Level and Premium Amount.



4.2 Predicting Existing Customer Segmentation

- Used the existing Segmentation Group as the target variable for a Random Forest classification model.
- Split the data into training and testing sets using an 80:20 ratio.
- Trained the model to predict the existing customer segmentation labels.
- Generated predictions on the test dataset.
- Reviewed feature importance to identify which numerical features contributed most to the model's predictions.





4.3 Future Purchase Prediction

- Examined available Purchase History dates to explore future purchase behavior.
- Created a prediction target based on whether a subsequent recorded purchase occurred within 30 days.
- Prepared the data for a future-purchase classification experiment.
- Evaluated the model using the available classification metrics and prediction outputs.
- Reviewed threshold-based predictions where applicable.

Important limitation: Customers without a later recorded purchase may not have enough information to determine whether they truly did not purchase again. The analysis therefore depends on the available purchase-history records and should not be interpreted as proof of actual future customer behavior.

```
Rows available for exploratory target: 2708

Balanced accuracy: 0.4914
Average precision: 0.0327
[[515   9]
 [ 18   0]]
              precision    recall  f1-score   support

           0       0.97      0.98      0.97       524
           1       0.00      0.00      0.00        18

    accuracy                           0.95       542
   macro avg       0.48      0.49      0.49       542
weighted avg       0.93      0.95      0.94       542
```

5. Model Evaluation and Validation

The machine learning models were evaluated using appropriate performance metrics.

- K-Means Evaluation: Used the silhouette score to assess the separation and consistency of the customer clusters.
- Classification Accuracy: Calculated accuracy for the Random Forest model predicting existing customer segmentation.
- Balanced Accuracy: Evaluated model performance while accounting for possible differences in class distribution.
- Classification Report: Reviewed precision, recall, and F1-score for the predicted customer groups.
- Confusion Matrix: Compared actual and predicted customer segmentation labels.
- Feature Importance: Examined the features contributing to the Random Forest model's predictions.
- Future Purchase Evaluation: Used balanced accuracy, average precision, and other available classification outputs to evaluate the future-purchase experiment.

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Segmentation model accuracy: 0.2575
Segmentation balanced accuracy: 0.2535

Validation checklist:
1. Review precision, recall, and F1-score for every class.
2. Compare the confusion matrix with the class distribution.
3. Check that target-derived columns are not used as input features.
4. Use a time-based split and a complete observation window for future-purchase prediction.
5. Do not rely on accuracy alone when classes are imbalanced.
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6. Business Insights and Marketing Recommendations

The customer segmentation results can help businesses develop more targeted marketing strategies. Final recommendations should be based on the actual characteristics of each cluster.

- High-Premium Customers:

- Identify customer groups with higher average premium amounts.
- Offer relevant insurance products and suitable coverage options.
- Provide personalized communication and customer support.

- High-Income Customers:

- Analyze customers with both high income and high premium amounts.

- Recommend suitable products based on customer preferences and existing ownership.
- Avoid assuming that high income alone guarantees purchase interest.

- Lower-Premium Customers:

- Explore affordable product options and suitable coverage plans.
- Use educational campaigns to explain product benefits.
- Review customer needs before offering discounts.

- Customer Segmentation:

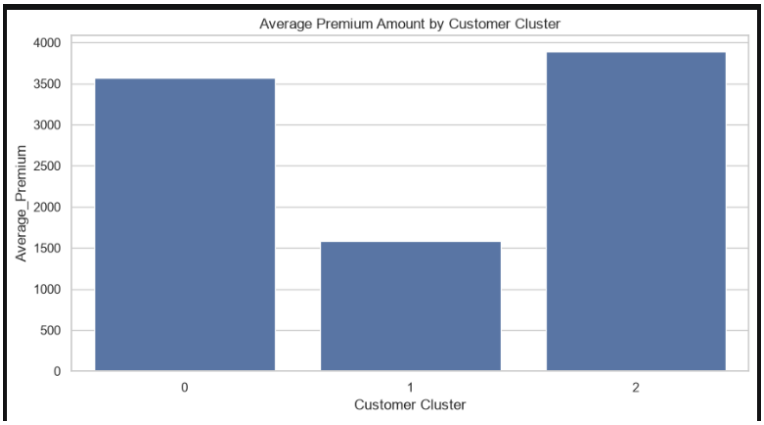
- Compare average income, premium amounts, coverage amounts, and other relevant characteristics across the three clusters.
- Design separate marketing campaigns for each group.
- Give each cluster a business-friendly label only after reviewing its actual characteristics.

- Purchase Behavior:

- Use available purchase-history information to understand recorded purchasing patterns.
- Test reminders and personalized offers where appropriate.
- Validate future-purchase predictions against subsequent real customer activity before relying on them for business decisions.

- Data Quality:

- Continue monitoring missing values, duplicate records, and inconsistent data.
- Ensure purchase-history dates are suitable for the intended prediction task.
- Avoid treating customers with no later recorded purchase as confirmed non-purchasers without sufficient observation data.



7. Tools and Technologies Used

- Python: Data cleaning, feature engineering, analysis, and machine learning.
- Pandas: Data manipulation, missing-value handling, and aggregation.
- NumPy: Numerical operations and data preparation.
- Matplotlib: Data visualization.
- Seaborn: Statistical plots and correlation heatmaps.
- Scikit-learn: Standardization, K-Means clustering, Random Forest classification, and model evaluation.
- PostgreSQL: Database integration code for storing customer data.
- SQLAlchemy: Connecting Python with PostgreSQL.

x8. Project Conclusion

The Customer Segmentation and Prediction project demonstrates a data science workflow involving data cleaning, exploratory data analysis, feature engineering, customer segmentation, and machine learning evaluation.

Python was used to prepare the customer dataset, analyze demographic and financial characteristics, and apply K-Means clustering to identify customer groups. A Random Forest classifier was used to predict existing customer segmentation labels, while a separate experiment explored future-purchase prediction using available purchase-history dates.

The resulting cluster summaries and model evaluations can help guide personalized marketing strategies, customer engagement, and product recommendations. However, final business conclusions should be based on actual model outputs and validated against real customer behavior.